



BLOCKCHAIN BASED GOVERNANCE

Although India is against cryptocurrencies, the government is exploring various use cases for adopting the underlying technology of blockchain

Aditya Madanapale | aditya@digit.in

While there have been many efforts at creating virtual currencies, the first real breakout success was that of Bitcoin. The

promise it offered was that the underlying blockchain, a decentralised distributed ledger system, made sure that no one could spend the same bitcoin, a digital asset, twice over. This could be guaranteed without recourse of an intermediary, such as a bank. Now blockchain has become so common that it has become synonymous with DLT*. The DLT is a new paradigm in data storage. In centralised data storage, the nodes in a system all connect to a central server. In decentralised storage, there may be many servers

to which some nodes have access. In distributed networks, there are no servers, and all the nodes connect to all the other nodes, forming a network. Within the blockchain, when a bitcoin transaction is requested, it is broadcast to the entire network, which validates it. A number of validated transactions are added together to form a new block in the distributed ledger. This block is added to the immutable blockchain. The concept can be extended to applications beyond cryptocurrencies - such as records, timestamps, event logs, health records, insurance, grants, and evidence books of the police. Using distributed ledger technologies (DLTs) for these applications ensures that the records cannot be fudged, are cryptographically secure, can be audited easily, and are resilient to single points of failures. The benefits are immense, cost and complexity is reduced with the removal of intermediaries, committing

frauds would be nearly impossible, and the documents will not be lost if a building burns up. Such a DLT would be always on, working around the clock. We look at four specific use cases when it comes to governance.

HEALTH RECORDS

In India, electronic health records are extremely fragmented, and are implemented through different models by institutions and organisations. Because of the various vocabularies used, a major problem is interoperability. Another problem is tracking the same patient across the different electronic health record systems. A centralised blockchain based electronic health record by the government could solve a lot of problems at once. Estonia has implemented just such a system. Every time, anyone in Estonia goes to a doctor, he is identified through an ID card (the Aadhaar card can be used here). The information